ECSE 331
Laboratory No. 4
Design of a BJT Amplifier in PSPICE

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***Abstract*—BJT's are a type of transistor that can be used to make amplifier circuits; where an input voltage is multiplied by a certain gain factor and an output voltage of the input multiplied by the gain is produced across a load. There are different templates of BJT amplifier circuits, where the gain factor depends on the resistor values of the circuit. To design a BJT amplifier circuit, the resistor values of a template BJT amplifier circuit must be determined based on a set of requirements. In this laboratory, a BJT amplifier circuit is to be designed while meeting a certain set of requirements. Thus, different resistor values of a chosen BJT circuit template must be found based on the given requirements. PSPICE circuit simulation software is to be used to test the circuit to see if the given requirements are met and to investigate temperature characteristics of the circuit.**

I. Introduction

This lab requires the design of a BJT amplifier circuit based on the following requirements:

- The circuit must be capable of operating off a $9V$ battery ($V_{CC} = 9V$)
- The voltage gain magnitude must be between $45^V/_V$ and $55^V/_V$ ($45^V/_V \leq \frac{V_{out}}{V_{in}} \leq 55^V/_V$)
- The circuit must drive a load of $10K\Omega$ ($R_L = 10K\Omega$)
- The circuit must have an input resistance of at least $1K\Omega$ ($R_{in} \geq 1K\Omega$)
- The $3dB$ bandwidth must be greater than $10KHz$

For the BJT amplifier circuit template, the common emitter amplifier without R_e is chosen. This circuit template can be seen in Figure 1. The resistor values of R_{sig}, R_1, R_2, R_C and R_E are found through theoretical means based on the given requirements. Also, the capacitor values of CC1, CC2 and CE are also found based on the given requirements.

To find the resistor values theoretically, both AC and DC analysis are performed, and a set of equations are derived. Also, several circuit parameter values are assumed to calculate the required resistances.

Once the values for the resistors and capacitors are determined theoretically, the circuit needs to be tested using circuit simulation software to verify the theoretical circuit parameter values. The circuit is constructed using PSPICE circuit simulation software and an AC sweep simulation is performed. By performing an AC sweep and plotting the gain of the circuit against frequency, the voltage gain of the circuit and the bandwidth are determined, and the parameter values are verified.

Finally, a temperature sweep simulation is performed on the amplifier circuit to examine the effects of temperature on the gain of the amplifier circuit. This is done by plotting the gain against frequency of the circuit at different temperatures.

II. Analysis

A. Theoretical Design

First, AC analysis is performed on the circuit, where a small signal model of the circuit is created. The small signal model, using the hybrid-π model, can be seen in Figure 5. Using the small signal model, an expression for the input resistance R_{in} is found in terms of R_1, R_2 and r_π. The expression for R_{in} can be seen in Equation 1 of Appendix A. Also, an expression for the voltage gain in terms of β , r_π, R_C and R_L is found and can be seen in Equation 6 of Appendix A. The small signal analysis and the full derivation of the expression for the voltage gain can be seen in Appendix A.

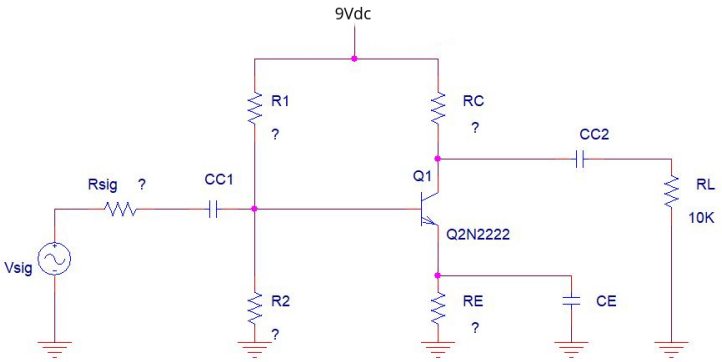


Fig. 1. Design template circuit schematic

Second, DC analysis is performed on the circuit; a DC model of the circuit is created. The DC model of the circuit can be seen in Figure 6. Using the DC model, an expression for V_B is found in terms of I_B, R_1, R_2 and VCC. The expression for the V_B can be seen in Equation 8 of Appendix B. The derivation of the expression for VB using the DC circuit is shown in Appendix B.

Next, values for R_1, R_2 and R_{sig} are chosen, as well as a value for β. The value for β is chosen to be 100 because β is often between 50 and 150 and 100 is an easy value to do calculations with. R_1 and R_2 are chosen to be $10K\Omega$ and R_{sig} is chosen to be $1K\Omega$. These values are chosen because most resistor values seen in class are in the thousands and the values chosen are easy to do calculations with.

Once values for R_1 and R_2 are chosen, a value for r_π is found using the expression for R_{in} derived using AC analysis. Based on the requirement that R_{in} has to be greater or equal to $1K\Omega$, the value for R_{in} is chosen to be $1K\Omega$. Now that values for R_1, R_2 and R_{in} are known, the value for r_π, using the R_{in} expression, is found to be $1.25K\Omega$. The calculations for r_π can be seen in Appendix B.

Now that the value for r_π is known, a value for R_C can be found using the expression for voltage gain, which was derived using AC analysis. Knowing that the value for $^{V_{out}}/_{V_{in}}$ must be between $45^V/_V$ and $55^V/_V$, the value for $^{V_{out}}/_{V_{in}}$ is chosen to be $50^V/_V$. Knowing that the value for R_L is required to be $10K\Omega$, the value for R_C can be found using the expression for voltage gain as values $^{V_{out}}/_{V_{in}}$, r_π and R_L are known. The value for R_C is found to be 666.67Ω. The relevant calculations for can be seen in Appendix B.

Next, the value for R_E is found. The process of finding R_E requires multiple calculation steps. The calculations for finding R_E can be seen in Appendix B.

1) Find the value of I_B. Using the expression for r_π in

terms of V_T and I_B, the value for I_B is found to be $2 \times 10^{-5} A$ since the values for r_π and V_T are known.

2) Use the expression for V_B, derived using DC analysis, to find the value for V_B now that the value for I_B is found. Since all values for the variables in the expression for V_B are known, the value for V_B is found to be $4.4V$.
3) Using V_B and I_B, values for V_E and I_E are found. Knowing that the value for V_E is $0.7V$ less than the value of V_B, the value of V_E is determined to be $3.7V$. Also, the value for I_E is determined to be $2.02 \times 10^{-3} A$ since I_E is equal to $I_B \times (\beta + 1)$.
4) Finally, knowing the values for V_E and I_E, the value for R_E is found by dividing V_E by I_E. The value of R_E is found to be 1831.68Ω.

B. Implementation of the design in PSPICE

The values of the capacitors are known be used to determine the bandwidth of the amplifier circuit. The capacitor values are most likely to be relatively large, therefore, relatively large capacitor values were chosen. Both CC1 and CC2 were chosen to have values of $10000F$ while CE was chosen to have a value of $0.000039F$. To determine whether the values of the capacitors are acceptable, a plot of voltage gain vs frequency was created to examine the bandwidth. The bandwidth was determined to be well over $10KHz$, thus satisfying the bandwidth requirement of the circuit as seen in Figure 3.

We Compile found values for unknown resistors and capacitors in Figure 1 for the BJT amplifier, into the circuit schematic shown in Figure 2 in PSPICE.

The correctness of our design can be determined by performing an AC sweep analysis on the amplifier circuit to determine the actual gain of the circuit. The gain can be seen by creating a graph of V_{out}/V_{in} vs frequency and observing which gain value the trace will reach. The plot of V_{out}/V_{in} vs frequency can be seen in Figure 3. From the plot, the gain is determined to be $46.9V/V$, which is within the required range for the gain. Therefore, the calculated resistor values are acceptable based on the gain requirement.

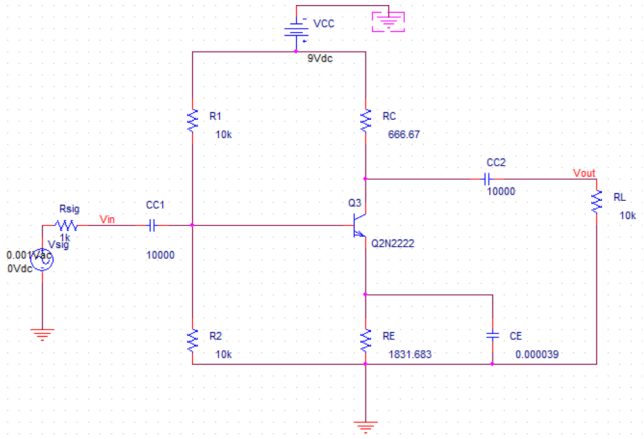


Fig. 2. BJT Amplifier design circuit schematic

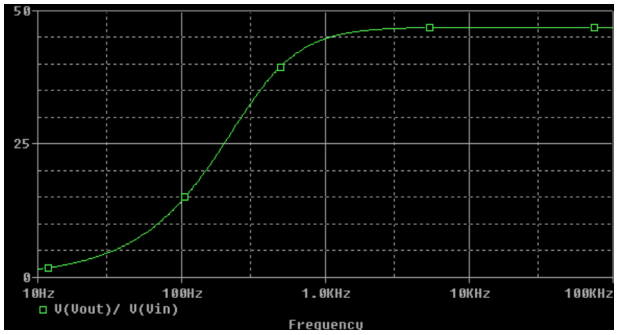


Fig. 3. BJT Amplifier V_{out}/V_{in} as a function of frequency

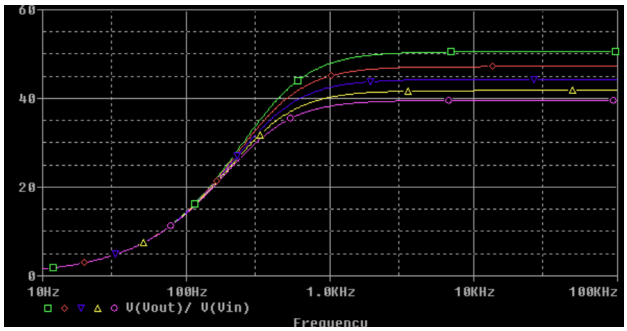


Fig. 4. BJT Amplifier V_{out}/V_{in} as a function of frequency at 0°C, 25°C, 50°C, 75°C, and 100°C

Next, an AC temperature sweep was performed on the amplifier circuit. The simulation was performed at temperatures of 0, 25, 50, 75 and 100 °C. A graph of voltage gain vs frequency at each of these temperatures is created so that the effect of temperature on the amplifier can be examined. The graph of voltage gain vs frequency at different temperatures can be seen in Figure 4. The traces of the gain at 0, 25, 50, 75 and 100 °C are represented by green, red, blue yellow and purple traces respectively. Based on the plot in Figure 4, the voltage gain decreased as temperature increased. The parameters of transistors change as the temperature changes, thus resulting in a change in the gain.

III. Conclusion

To conclude, in this laboratory, we designed a BJT amplifier given specific requirements. This design exercise required us to design a template circuit and find unknown circuit component values such as resistors and capacitors. We found circuit components by performing theoretical calculations using the small signal model. We then modelled the circuit in PSPICE and performed simulations to validate our design. We followed by studying the temperature characteristics of the designed amplifier. We concluded this design project by finding an inverse proportionality relation between temperature and amplifier voltage gain.

IV. Appendices

We need to find resistor values for R_1, R_2, R_{sig}, R_E, and R_C in Figure 5.

A. Small Signal Model (AC Analysis)

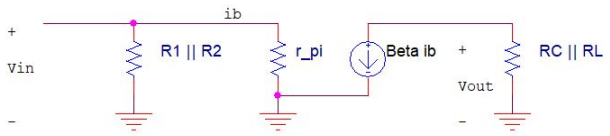


Fig. 5. Small signal model (AC Analysis) circuit

From the small signal model we can deduce:

$$Rin = R_1||R_2||r_\pi \tag{1}$$

$r_\pi = {}^{V_T}/_{I_B}$ and $V_T = 25 \times 10^{-3}V$ therefore:

$$I_B = \frac{V_T}{r_\pi} = \frac{25 \times 10^{-3}}{r_\pi} \tag{2}$$

Write V_{out} as a function of i_b and i_b as a function of V_{in}:

$$V_{out} = -\beta i_b (R_C||R_L) \tag{3}$$

$$i_b = {}^{V_{in}}/_{r_\pi} \tag{4}$$

From (3) and (4) we can write V_{out} as follows:

$$V_{out} = -\beta \frac{V_{in}}{r_\pi}(R_C||R_L) \tag{5}$$

Rewriting Equation (5) we get the following:

$$\frac{V_{out}}{V_{in}} = \left|\frac{-\beta}{r_\pi}(R_C||R_L)\right| \Rightarrow \frac{V_{out}}{V_{in}} = \frac{\beta}{r_\pi}(R_C||R_L) \tag{6}$$

B. (Small Signal Model) DC Analysis

$$V_B - V_E = 0.7V \tag{7}$$

$$\frac{V_{CC} - V_B}{R_1} = I_B + \frac{V_B}{R_2} \Rightarrow V_B = \frac{-I_B R_1 R_2 - V_{CC} R_2}{R_1 + R_2} \tag{8}$$

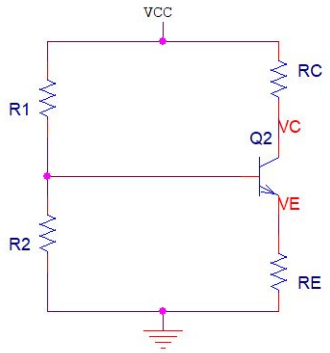


Fig. 6. DC Analysis circuit

Choosing the following parameters: $\begin{cases} R_1 = R_2 = 10K\Omega \\ R_{sig} = 1K\Omega \\ R_{in} = 1K\Omega \\ \beta = 100 \end{cases}$

and From Equation (1) we find r_π:

$$R_{in} = R_1||R_2||r_\pi = 1K\Omega \Rightarrow r_\pi = 1250\Omega$$

From Equation (2) we find I_B:

$$I_B = \frac{25 \times 10^{-3}}{1250} = 2 \times 10^{-5}A$$

Finding R_C using (6) and knowing that $R_L = 10K\Omega$:

$$\frac{V_{out}}{V_{in}} = \frac{100(R_C||R_L)}{r_\pi} = 50 \Rightarrow \boxed{R_C = 666.67\Omega}$$

Using Equation (8) with $V_{CC} = 9V$, we find $V_B = 4.4V$ so from (7), we get $V_E = 3.7V$.

$$I_E = (\beta + 1)I_B \Rightarrow I_E = 2.02 \times 10^{-3}A$$

$$R_E = \frac{V_E}{I_E} \Rightarrow \boxed{R_E = 1831.68\Omega}$$

Resistor values to be used in our design:

$$\begin{cases} R_1 = 10K\Omega \\ R_2 = 10K\Omega \\ R_{sig} = 1K\Omega \\ RC = 666.67\Omega \\ R_E = 1831.68\Omega \end{cases}$$